VIT-AP UNIVERSITY

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-32

Faculty

Dr. Rajasekhar N V

Associate Professor G-1

SENSE, VIT-AP University

Mail: rajasekhar.venkata@vitap.ac.in

Cabin: Faculty Area 424-C:AB-1

Outline of Today's Class

- BJT Introduction
- BJT with fixed bias
- Problems

Bipolar Junction Transistor(BJT) Construction

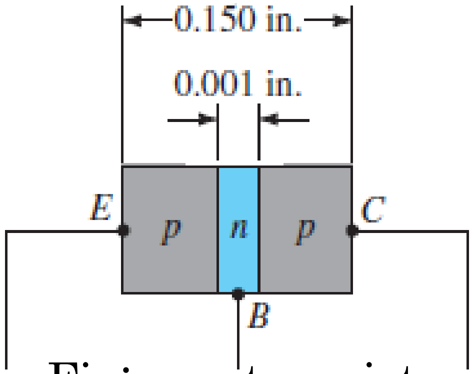


Fig: pnp transistor

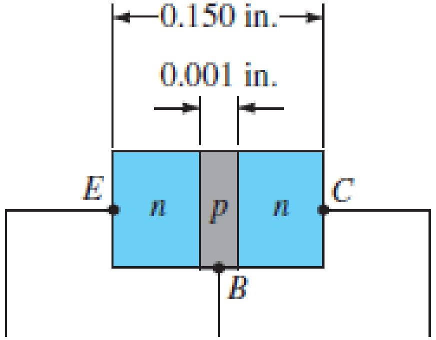
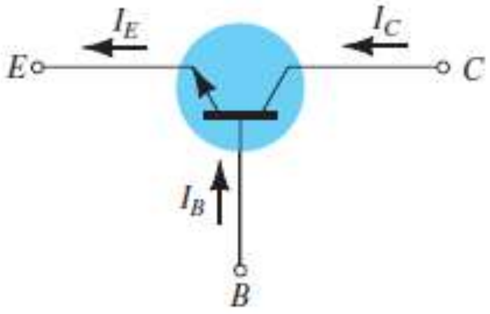
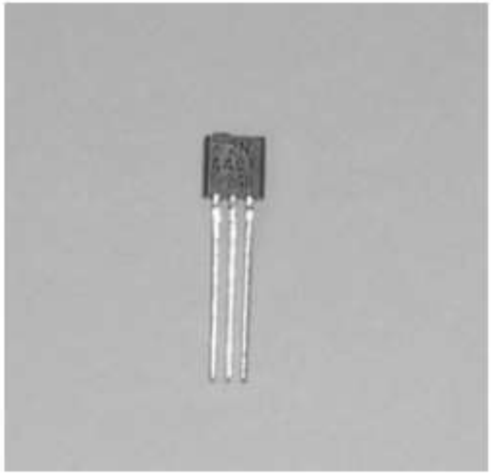
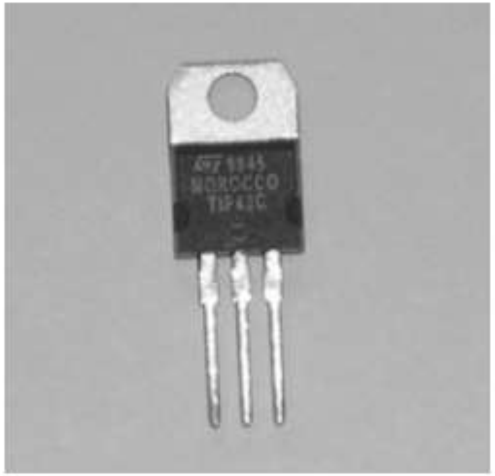


Fig: npn transistor





(a)



(b)

(c)

Various types of general-purpose or switching transistors: (a) low power; (b) medium power; (c) medium to high power.

Bipolar Junction Transistor(BJT) Construction

- The transistor is a three-layer semiconductor device consisting of either two n-type and one p -type layers of material or two p type and one n -type layers of material.

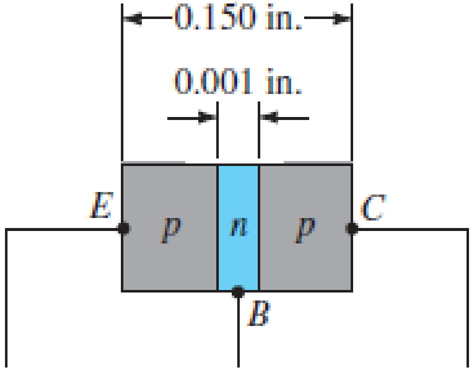


Fig: pnp transistor

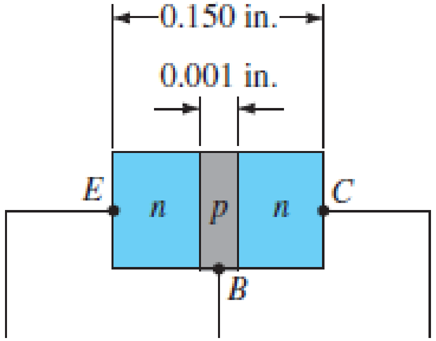


Fig: npn transistor

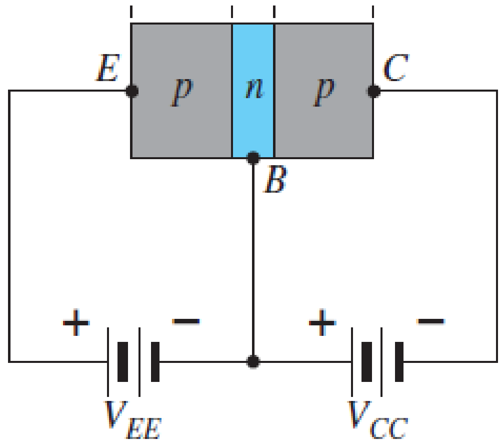
- *E* for *emitter* ,
- *C* for *collector* ,
- *B* for *base* .

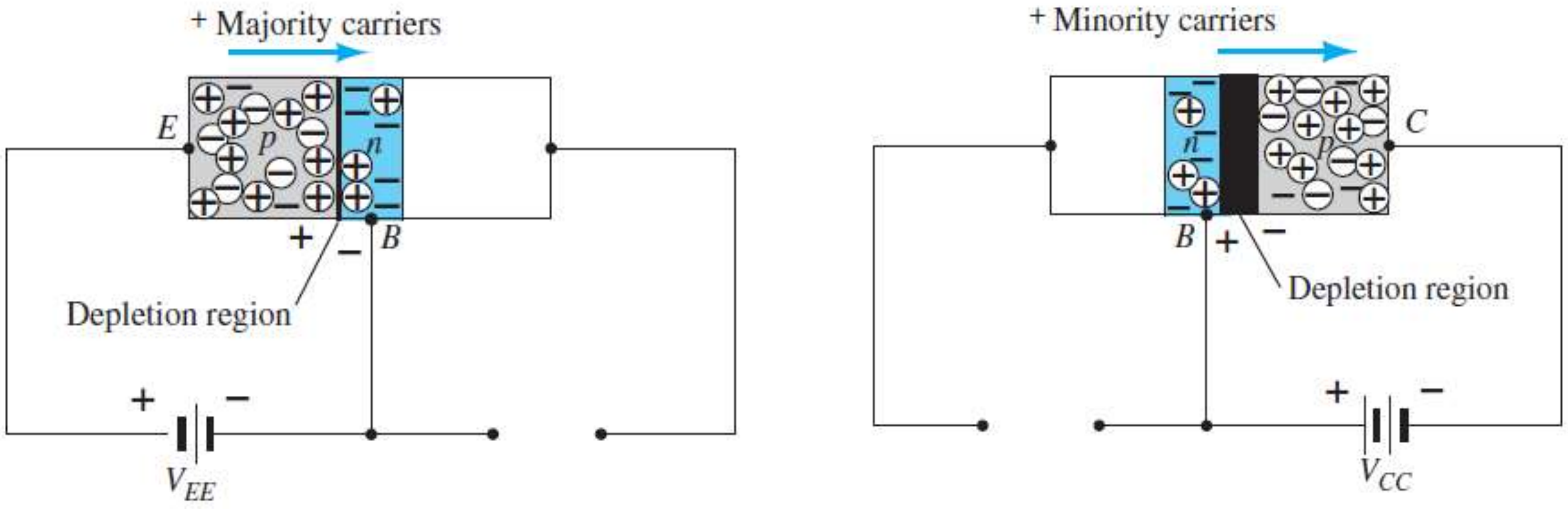
- The emitter layer is heavily doped, with the base and collector only lightly doped.
- The outer layers have widths much greater than the sandwiched *p* - or *n* -type material.

- The abbreviation BJT, from bipolar junction transistor , is often applied to this three-terminal device. The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material.
- If only one carrier is employed (electron or hole), it is considered a unipolar device.

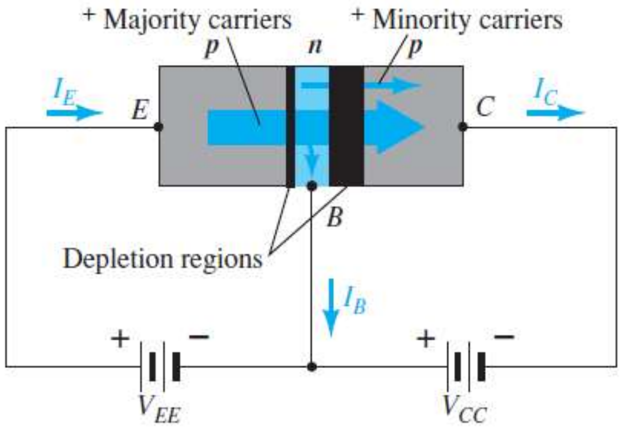
Types of transistors: (a) pnp (b) npn.

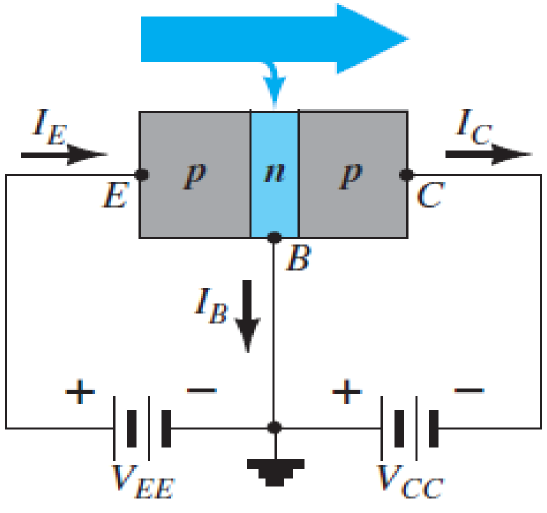
Transistor Operation



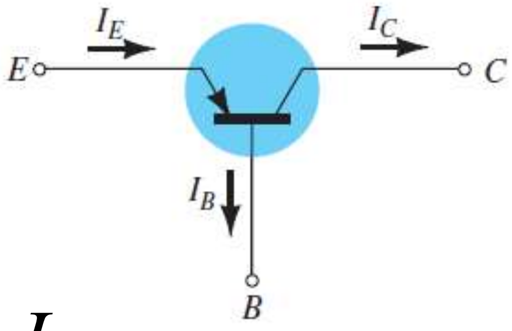


Biasing a transistor: (a) forward-bias; (b) reverse-bias.



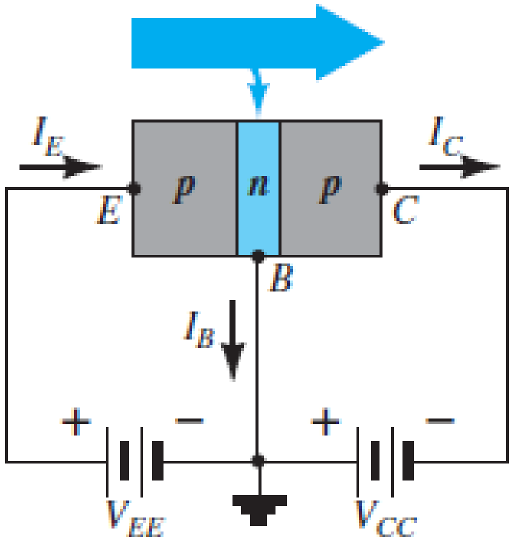


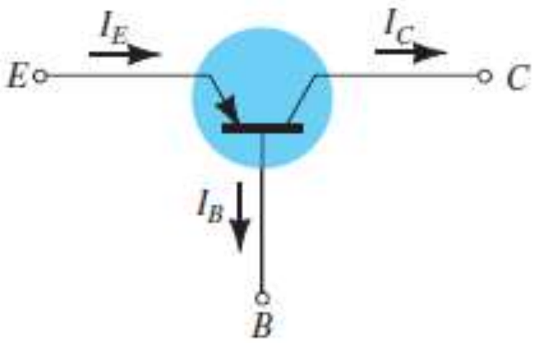
$$I_E = I_C + I_B$$



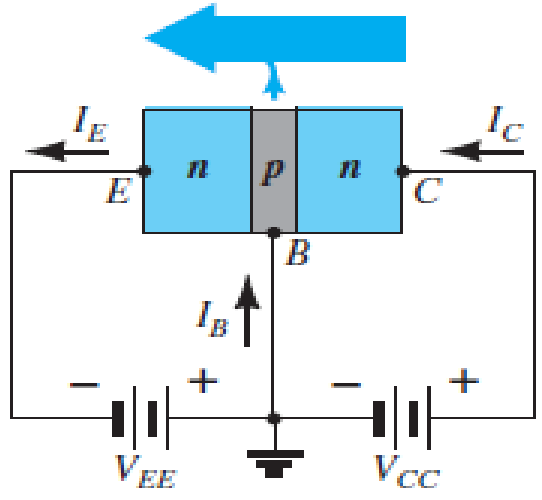
$$I_C = I_{C\text{ majority}} + I_{CO\text{ minority}}$$

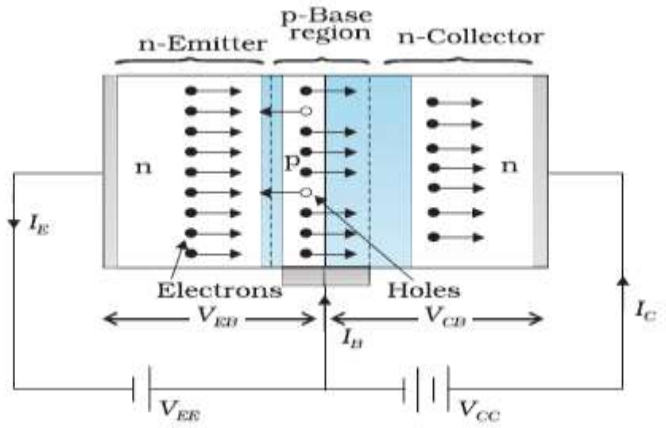
pnp

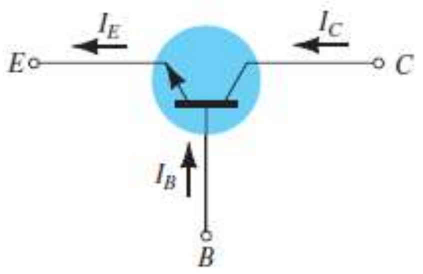




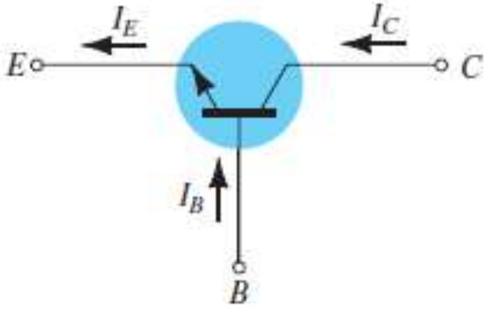
npn

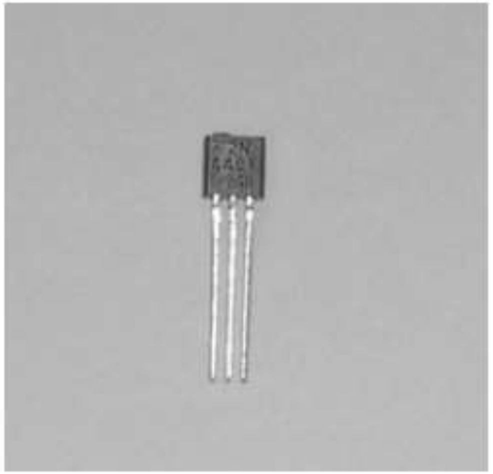




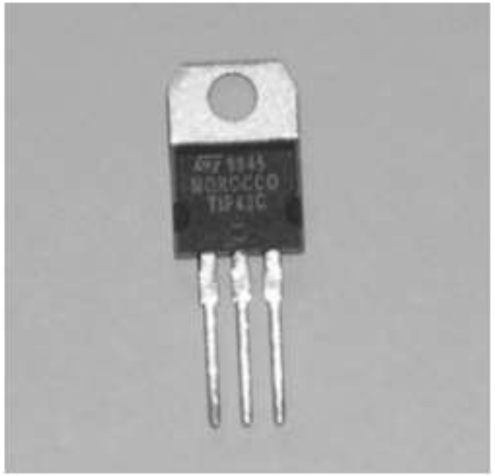


$I_E = I_C + I_B$





(a)



(b)

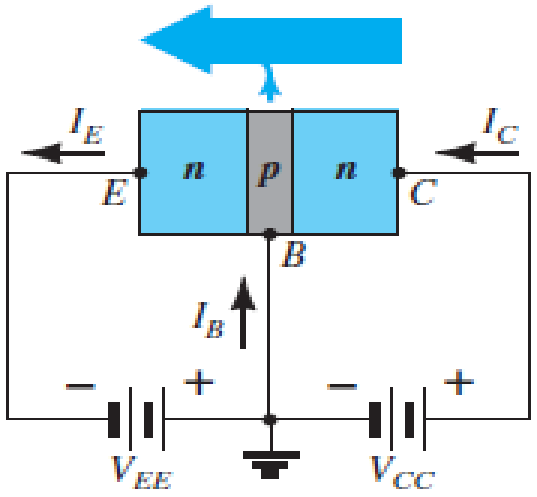
(c)

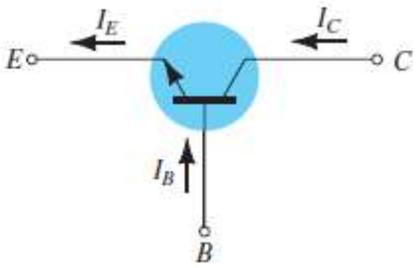
Various types of general-purpose or switching transistors: (a) low power; (b) medium power; (c) medium to high power.

BJT Configurations

1. Common-Base Configuration (CB)
2. Common-Emitter Configuration(CE)
3. Common-Collector Configuration(CC)

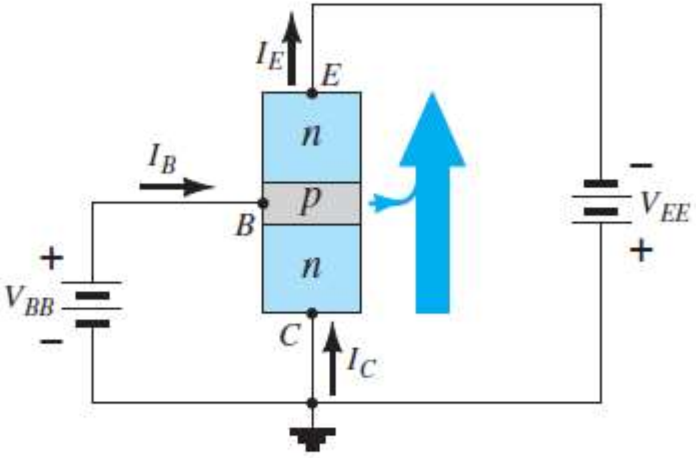
1.Common-Base(CB)

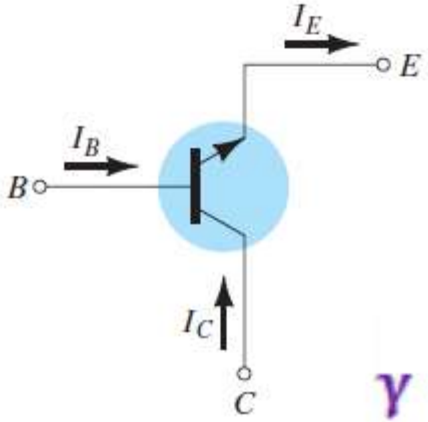




$$\alpha_{dc} = \frac{I_C}{I_E}$$

2.Common-Collector

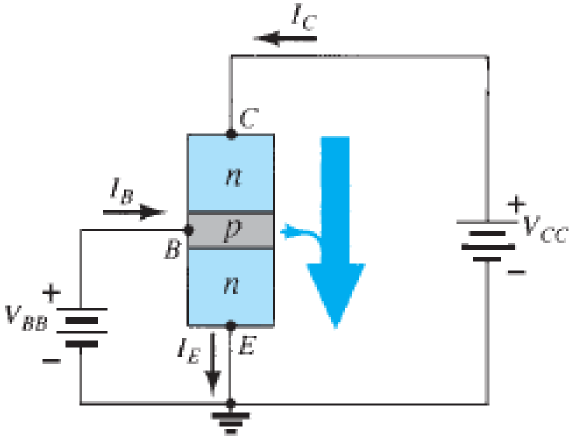


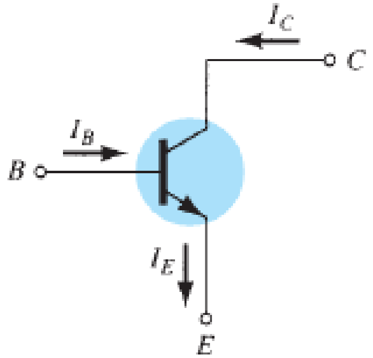


$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

$I_E = I_C + I_B$

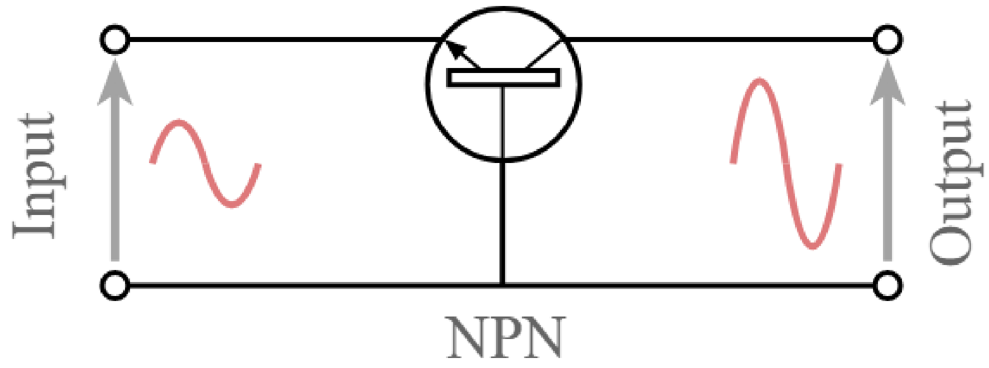
3.Common-Emitter





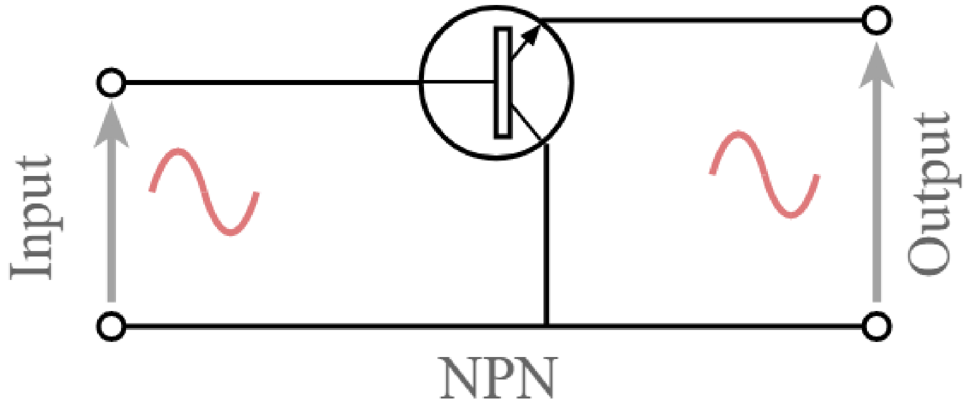
$$\beta_{dc} = \frac{I_C}{I_B}$$

1. Common-Base Configuration (CB)



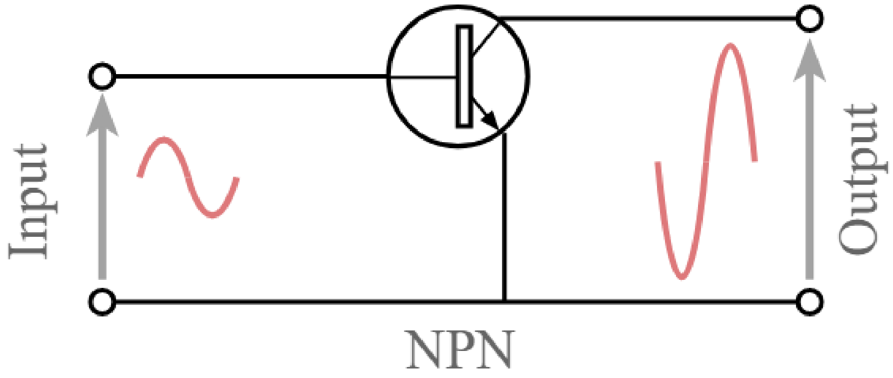
$$\alpha_{dc} = \frac{I_C}{I_E}$$

2. Common-Collector Configuration (CC)



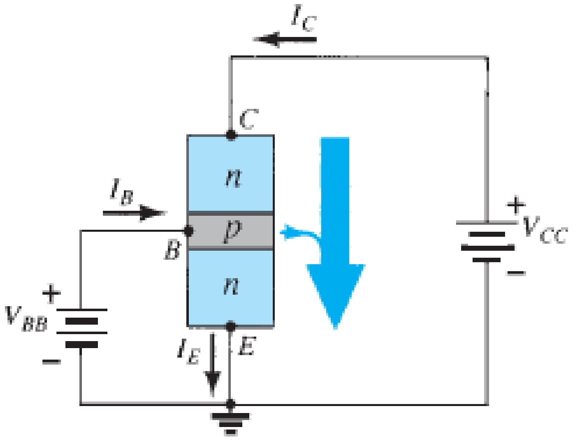
$$\gamma = \frac{I_E}{I_B}$$

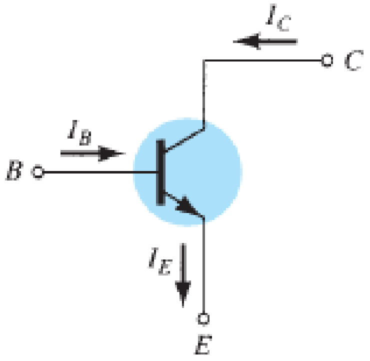
3. Common-Emitter Configuration (CE)

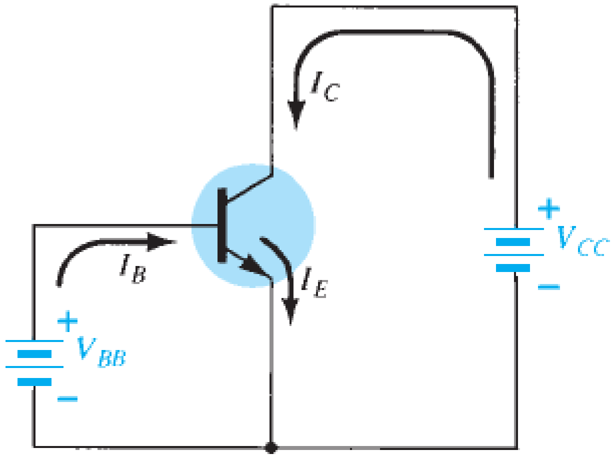


$$\beta_{dc} = \frac{I_C}{I_B}$$

3.Common-Emitter Configuration(CE)







$$\boxed{\beta_{dc} = \frac{I_C}{I_B}}$$

$$\boxed{I_C = \beta I_B}$$

$$\begin{aligned} I_E &= I_C + I_B \\ &= \beta I_B + I_B \end{aligned}$$

$$\boxed{I_E = (\beta + 1) I_B \cong I_C}$$